

Jay Dhakan

Python & AI/ML Engineer

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Python & AI/ML Engineer (2+ yrs) who ships **production-grade** data pipelines, **LLM/RAG chatbots** and **multi-agent automation** at scale. Cut a nightly ETL from 18 h → 1.2 h, halved third-party API spend, and **tech lead a 4-dev team** that delivers pixel-perfect results and 99.9 % uptime. **Youngest winner** of the **2024 Mr Perfectionist award** for quality and on-time delivery.

Experience

- **Python & AI/ML Engineer - BotPro Solutions - AI Chatbots** 2023 - present
 - **Lead a 4-person platform squad**; run **architecture reviews**, **sprints** and **hiring interviews**.
 - **Performance increase**: rebuilt legacy job with **multiprocessing** + **concurrent.futures**, shrinking runtime **17 h → ~80 min** and freeing **16 CPU-hours per run**.
 - **Optimized code** & calls to 3rd Party tools & services, **cutting SaaS cost 50 % (>\$700 / mo)**.
 - Deployed **GPU-accelerated image-similarity micro-service** (OpenCV + Siamese CNN) that spots matches in < 200 ms across a stored images.
 - Launched **1-click WhatsApp/Web chatbot creator** on OpenAI Assistants API—scrapes or ingests training data and deploys in < 5 min.
 - Set up and looked after **multiple production servers**, rolling out new services and keeping them running smoothly.
 - Conducted **10+ technical sessions** and mentored juniors to promotion track.

Projects

- **Zillow Data Harvester** 2023
 - **Python 3.11 + Scrapy + MongoDB**; **proxy rotation**, dynamic UAs, **smart throttling**
 - **100 + req/sec → ~ 70 M records/1.4 TB** harvested; full market sweep in 3 months
 - **150+ data points** per property (price history, images, zestimates, dimensions)
 - Weekly delta-crawl keeps dataset fresh; dashboards expose trends & crawler KPIs
 - Runs on lean infra—cost per million rows < \$3
 - **Smart Scheduler**: autorun the required services to move on to next phase when a full iteration is completed
- **Multi-Agent Prompt-to-Site Studio** 2025
 - **Next.js & Vercel front-ends** driven by OpenAI APIs and using **agency swarm** framework.
 - **4-5 min/site**; 90% dev-time & 75 % cost savings
 - **Separate agents** to handle niche **research**, **content generation**, **code**, **QA** & **Deployment** autonomously
 - Built a **multi-model stack**: **o4-mini-high** writes the code, GPT-4.1 lets the agents talk to each other, and **GPT-4.5(Manager Agent)** runs the whole workflow.
 - **Prompt Engineering**: Wrote the prompts and **rules each agent follows**, so they **stay on track** and give the right output.
 - Firebase to store links for sites with products; **GeniusLink API** optimizes multi-merchant routing
 - **99.9% deployment success**, zero-downtime rollouts

Projects

● Social Media Scraper

2024

- Tracks **1 500 accounts** across **6 social networks**; **20K new posts/day**
- **Master-Slave Architecture**: Re-engineered to 100% async a **Collector API (master)** streams crawl/refresh/download jobs, letting **worker pods** pull tasks on demand.
- Master-slave micro-cluster (**Collector ↔ Crawler / Refresher / Downloader**) auto-scales on server Spot.
- **Rotating proxies** from multiple providers + **adaptive retry** and **exponential back-offs** = **99% success** with zero downtime
- **20 + REST endpoints** for **on-demand** downloads, crawls, refreshes, health & analytics
- **Realtime ops custom dashboard**: plots **per-platform KPIs**—posts crawled / refreshed / downloaded—with 24 h and 30-day trendlines for instant visibility.
- **Fleet-health panel**: **tracks 40+** crawler / refresher / downloader pods; monthly Query roll-ups list accounts and media processed, **flagging any lagging worker**.
- **Smart alerting**: structured logs d that ping **Slack & MSTEams** on stalled queues or error spikes, slashing silent data-loss incidents by 90 %.

● Custom Google Search Kit

2025

- **FastAPI + HTTPX async** workers behind **Gunicorn/Uvicorn**; Redis for **rate-limit** store
- End-to-end latency stays **< 800 ms per request**, during peak load (**<2 s**).
- Benchmarked at **175+QPS** with **95+ % success** under fault-injection tests
- Returns **10+ structured fields** per result (title, URL, snippet ...)
- Pluggable back-ends (Bing, DuckDuckGo) swap in with zero downtime
- Basic-Auth & **per-API key limits provide SaaS-ready tenant isolation**
- FastAPI + Uvicorn with keep-alive pools, payloads, pushes **15M+ queries/day** through a single **4-vCPU, 8 GB node**, while CPU stays < 70 %.
- **Usage analytics**: nightly Query jobs roll up—**top keywords**, and **per-key consumption**—to feed billing and capacity planning.

Education

Marawadi University

2019-2023

- Bachelor of Technology, Specialized in Artificial Intelligence
- CGPA: 8.02

Tech Stack

- **Flask, FastAPI**
- **MongoDB**, AzureDB, PostgreSQL, DynamoDB, SupdaBase
- Pandas, PySpark, Numpy, Matplotlib
- **PyTorch, OpenCV**
- GitHub(CI/CD), **Docker**, **pm2**
- **Pants** Build System
- RAG, LLM, llamaindex, LangChain/LangGraph
- VectorDB **FAISS** | **Pinecone** | **Qdrant** | **Chroma**
- **AWS**, S3, ec2, contabo, CloudWatch
- **Scrapy**, BeautifulSoup4, Selenium
- **Playwright**, Requests, **AioHTTP**, **HTTPX**
- **APScheduler**, os, Redis
- **Tkinter**, **nlTK**, regex
- Logging, **Pydantic**, marshmallow
- agency-swarm, GroqAI, OpenAI APIs
- hugging-face, transformers